

MOSJOEN KJAERSTAD, NorwayWMO: **011220**Lat: **65.784N**Lon: **13.218E**Elev: **72**StdP: **100.46**Time Zone: **1.00 (EUC)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.2	-23.9	0.4	-20.5	-21.1	0.6	-17.2	10.4	-0.4	9.2	-0.4	1.8	170	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	25.2	16.4	22.8	15.3	20.8	14.3	17.6	23.1	16.3	21.2	15.1	19.4	3.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	19.9	14.1	10.1	18.3	13.0	9.4	17.0	49.7	23.1	45.8	21.4	42.4	19.6	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.9	5.9	DB	-22.1	28.6	3.0	2.8	-24.3	30.6	-26.0	32.2	-27.7	33.8	-29.9	35.9	(4)
(5)			WB	-22.3	18.8	2.8	1.7	-24.3	20.0	-26.0	21.0	-27.5	21.9	-29.5	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-5.4	-4.5	-1.7	2.9	7.6	11.0	14.5	13.0	9.8	4.0	-1.3	-3.5	(6)	
	DBStd	8.24	6.35	6.33	4.75	2.94	3.84	3.42	3.51	2.66	3.01	3.77	5.59	6.26	(7)	
	HDD10.0	2571	476	406	362	214	98	27	1	4	39	188	338	419	(8)	
	HDD18.3	5274	734	639	621	464	333	223	129	166	256	444	588	677	(9)	
	CDD10.0	352	0	0	0	0	23	56	141	97	33	2	0	0	(10)	
	CDD18.3	14	0	0	0	0	0	2	11	1	0	0	0	0	(11)	
	CDH23.3	179	0	0	0	0	12	17	132	17	1	0	0	0	(12)	
	CDH26.7	41	0	0	0	0	1	3	36	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.3	2.4	2.7	2.4	2.5	2.6	2.3	2.0	2.0	2.2	2.1	2.3	(14)	
(15) Precipitation	PrecAvg	1685	182	173	172	110	79	84	87	94	166	179	158	200	(15)	
	PrecMax	2316	394	376	326	253	127	194	213	190	471	362	322	375	(16)	
	PrecMin	1192	15	23	33	30	25	19	36	21	44	62	14	35	(17)	
	PrecStd	262	98	99	88	60	24	43	42	36	90	78	88	101	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	7.5	8.1	17.2	25.2	25.7	30.8	25.4	21.2	15.2	10.3	8.2	(19)	
		MCWB	6.4	5.5	4.7	8.9	13.5	15.7	19.4	16.4	13.1	10.5	8.8	6.8	(20)	
	2%	DB	5.2	6.2	6.9	12.1	21.4	22.1	26.9	22.9	18.4	13.0	8.9	6.9	(21)	
		MCWB	3.5	4.6	4.6	6.2	12.5	13.9	17.7	15.6	12.2	8.7	7.3	5.4	(22)	
	5%	DB	3.7	5.1	5.7	9.5	18.8	20.0	24.0	20.8	16.7	11.2	6.5	5.0	(23)	
		MCWB	2.4	3.5	3.5	5.1	11.5	13.0	16.5	14.7	11.5	8.5	4.8	3.3	(24)	
	10%	DB	2.1	3.6	4.6	7.8	15.0	17.9	21.5	18.8	14.9	9.5	4.8	3.7	(25)	
		MCWB	1.2	2.0	2.6	4.3	9.7	12.1	15.3	13.9	11.3	7.5	3.5	2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.7	6.0	9.2	14.6	16.9	20.2	18.1	15.0	10.9	9.3	7.0	(27)	
		MCDB	7.5	7.0	7.4	16.5	23.0	24.4	27.2	22.4	18.6	14.1	10.3	8.0	(28)	
	2%	WB	3.8	4.8	4.7	7.1	13.1	15.2	18.6	16.5	13.7	9.7	7.1	5.6	(29)	
		MCDB	4.8	6.2	6.1	10.9	20.0	20.5	25.0	21.2	16.8	12.0	8.7	6.7	(30)	
	5%	WB	2.3	3.7	3.7	5.8	11.7	13.9	17.3	15.4	12.6	8.6	5.1	3.4	(31)	
		MCDB	3.3	5.0	5.3	8.2	17.9	18.5	22.8	19.6	15.3	10.8	6.1	4.5	(32)	
	10%	WB	1.2	2.2	2.7	4.8	9.9	12.7	15.9	14.4	11.6	7.7	3.7	2.3	(33)	
		MCDB	2.0	3.3	4.4	7.1	14.9	16.9	20.6	18.2	14.0	9.4	4.6	3.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.4	7.7	8.0	9.1	7.9	8.4	9.1	8.2	6.1	4.9	5.5	(35)	
		MCDBR	5.3	6.0	6.6	11.5	15.1	12.9	13.6	13.6	11.6	7.4	5.8	5.8	(36)	
	5% WB	MCWBR	4.5	4.9	4.8	6.7	7.3	6.2	5.8	7.0	6.3	5.1	4.7	4.7	(37)	
		MCWBR	5.1	6.1	5.9	9.3	14.0	11.4	12.1	11.9	9.3	6.5	5.8	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.278	0.311	0.338	0.343	0.343	0.352	0.338	0.321	0.295	0.353	N/A	(40)	
		taud	N/A	2.142	2.196	2.243	2.363	2.473	2.475	2.505	2.535	2.353	2.615	N/A	(41)	
	Ebn at Noon		N/A	580	746	822	859	870	848	827	760	576	241	N/A	(42)	
		Edh at Noon	N/A	55	85	104	102	94	92	81	63	45	19	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.08	0.45	1.72	3.38	4.47	4.82	4.46	3.46	1.84	0.73	0.15	0.02	(44)	
	RadStd		0.01	0.09	0.27	0.37	0.32	0.60	0.53	0.38	0.26	0.13	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	-93	-133	N/A	N/A

Nomenclature: See separate page